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Correct Answer

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Practice Exam - 1

Question 54 of 117



A 51-year-old woman is being evaluated for gradually progressive dyspnea on exertion over the past year. She has no known prior history of hypertension, cardiac or lung disease. She was never a smoker. She is not on any medications. She now has to stop after walking half a block and can no longer climb stairs in her home without resting periodically.

On examination, her blood pressure is 120/80 mm Hg, heart rate is 90 bpm, respiratory rate is 16 breaths/ min, and oxygen saturation is 93% RA. Her jugular venous pressure is 8 cm H₂O. Cardiac examination is unremarkable except for a widely split S₂ and loud P₂. Lungs are clear to auscultation. There is 1+ bilateral pedal edema.

An echocardiogram shows normal left ventricular ejection fraction, but a moderately dilated right ventricle (RV) with moderate RV systolic dysfunction. The interventricular septum is D-shaped. The pulmonary artery systolic pressure is estimated at 50 mm Hg. She is referred for a ventilation/perfusion (V/Q) scan, which demonstrates bilateral segmental mismatched perfusion defects. The patient is started on therapeutic anticoagulation and then referred for definitive surgical management.

Which one of the following preoperative hemodynamic parameters is associated with the highest risk of mortality following pulmonary thromboendarterectomy?

A. Central venous pressure (CVP)/Pulmonary capillary wedge pressure (PCWP) ratio of 0.6

B. Mean pulmonary artery pressure of 50 mm Hg

C. Pulmonary vascular resistance of 1200 dyn s cm⁻⁵

D. Pulmonary artery pulsatility index of 1.9

E. Right ventricular stroke work index of 400

Testing Point: The Board-certified AHFTC specialist should know the risk factors for mortality with respect to definitive surgical management of WHO Group 4 chronic thromboembolic pulmonary hypertension.

Correct Answer: C. Pulmonary vascular resistance of $1200 \text{ dyn}\cdot\text{s}\cdot\text{cm}^{-5}$.

Rationale: Based on the clinical findings of severe pulmonary hypertension with a V/Q scan demonstrating bilateral pulmonary emboli, the patient has chronic thromboembolic pulmonary hypertension. Pulmonary thromboendarterectomy (PTE) offers the best chance at symptomatic improvement and long-term survival in eligible patients. At experienced centers, 1-year reported survival post-PTE is $> 90\%$, 5-year survival is $> 80\%$ and 10-year survival is $> 70\%$. Even at experienced centers, however, perioperative mortality is increased with markedly elevated preoperative pulmonary vascular resistance (PVR) $> 1000 \text{ dyn}\cdot\text{s}\cdot\text{cm}^{-5}$, corresponding to > 12.5 Wood Units. At the UCSD center, 4.1% of patients with a pre-operative PVR $> 1000 \text{ dyn}\cdot\text{s}\cdot\text{cm}^{-5}$ died, whereas only 1.6% who had a PVR $< 1000 \text{ dyn}\cdot\text{s}\cdot\text{cm}^{-5}$ died. Data from the international CTEPH registry showed in-hospital mortality approximately three times greater in those with PVR $> 1200 \text{ dyn}\cdot\text{s}\cdot\text{cm}^{-5}$ at diagnosis compared with PVR $400\text{--}800 \text{ dyn}\cdot\text{s}\cdot\text{cm}^{-5}$. Although the other answer choices represent markers of increased postsurgical mortality risk, the markedly elevated PVR shown here has been demonstrated to confer the highest risk.

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